

picopdf

A minimalist markdown to PDF converter with custom font support.

Building

```
cargo build --release
```

The binary will be at `target/release/picopdf`.

Usage

Convert markdown to PDF using default fonts:

```
picopdf write -i input.md -o output.pdf
```

Use custom TrueType fonts:

```
picopdf write -i input.md -o output.pdf \ --font-regular ~/fonts/regular.ttf \  
--font-bold ~/fonts/bold.ttf \ --font-mono ~/fonts/mono.ttf
```

Architecture

Four-stage pipeline:

- Markdown to AST/IR using pulldown-cmark
- AST to styled blocks with semantic styling
- Blocks to positioned lines and glyphs via layout engine
- Lines to PDF using pdf-writer with embedded fonts

Font Support

Default fonts are built-in PDF Type1 fonts (Helvetica, Courier) requiring no embedding. Custom TrueType/OpenType fonts are fully embedded in the PDF with proper Unicode support.

Default PDFs are around 2KB. Custom font PDFs include the full font files, typically 2-7MB depending on font complexity.

Supported Markdown

- Headings (H1-H4)
- Paragraphs
- Lists
- Code blocks
- Inline code

Example

```
# Using 0xProto Nerd Font picopdf write -i README.md -o README.pdf \  
--font-regular ~/.local/share/fonts/0xProto-Regular.ttf \ --font-bold \  
~/.local/share/fonts/0xProto-Bold.ttf \ --font-mono \  
~/.local/share/fonts/0xProtoMono-Regular.ttf
```